



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Department of Computer Science and Engineering DevOps Lab - L3

Lab Assignment No: 2	
Name of Student	Raghav Dhoot
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Batch	2023 - 2027
Class	CSE - B1
Academic Year & Semester	2026 - 2027, 7th Semester
Title of Assignment	Git Branching & Merging
Implementation & Theory	<pre>● (base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework % git status On branch main Your branch is up to date with 'origin/main'. nothing to commit, working tree clean</pre> git status (initial check) — Confirmed main was clean and up to date with origin/main before starting. <pre>● (base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework % git checkout -b feature-update Switched to a new branch 'feature-update' ● (base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework % git branch * feature-update main</pre> git checkout -b feature-update + git branch — Created and switched to a new branch feature-update, confirmed with git branch showing it as the active branch. <pre>● (base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework % git status On branch feature-update Changes not staged for commit: (use "git add <file>..." to update what will be committed) (use "git restore <file>..." to discard changes in working directory) modified: README.md no changes added to commit (use "git add" and/or "git commit -a")</pre> nano + git status (modified) — Edited README.md to mark assignment 01 as "Completed," git status shows it as modified but unstaged.

```

• (base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework % git add .
• (base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework % git status
On branch feature-update
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   README.md
• (base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework % git commit -m "Update README for Assignment 2"

```

git add + git commit + git status — Staged and committed the change on feature-update with message "Update README on feature branch." git status confirms a clean tree after commit.

```

• (base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework % cat README.md
# DevOps Coursework

This repository contains the DevOps coursework assignments.

## Assignment 1

Git Setup and Basic Workflow. ✅ Completed

## Assignment 2

Git Branching and Merging. 🔄

```

git checkout main + cat README.md — Switched back to main and printed README.md, showing the "Completed" edit is absent — proves the change is isolated to the feature branch.

```

• (base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework % git merge feature-update
Updating aa39d7c..e7052fc
Fast-forward
 README.md | 6 +++++
 1 file changed, 5 insertions(+), 1 deletion(-)

```

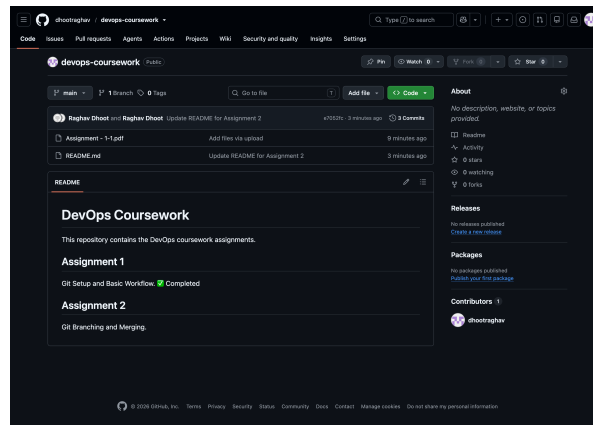
git merge feature-update — Merged feature-update into main; output shows a fast-forward merge. cat README.md afterward confirms the change is now present on main.

```

• (base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework % git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 404 bytes | 404.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:dhootraghav/devops-coursework.git
 aa39d7c..e7052fc  main -> main

```

git push origin main — Pushed the updated main branch to GitHub, reflecting the merged commit.



Repo on GitHub after push — Verified on GitHub that README.md shows assignment 01 marked "Completed," confirming the merge was pushed successfully.